1 Commutativity

$$(f*g)(x) = \int_{\mathbb{R}^d} f(y)\, g(x-y)\, dy$$

$$z = x - y$$

$$(f*g)(x) = \underbrace{\int_{\mathbb{R}^d} f(x-z)\, g(z)\, dz}_{g*f}$$

$$\Rightarrow \boxed{f*g = g*f}$$

2.

$$[f*(g*h)](x) = \iint f(y)\, g(z)\, h(x-y-z)\, dz\, dy$$

$$t = y+z$$

$$\int \underbrace{\left[\int f(y)\, g(t-y)\, dy\right]}_{(f*g)(t)} h(x-t)\, dt$$

$$\int (f*g)(t)\, h(x-t)\, dt = [(f*g)*h](x)$$

$$\Rightarrow f*(g*h) = (f*g)*h \quad \checkmark$$

3.

$$[f*(g+h)](x) = \int f(y)\,(g(x-y)+h(x-y))\, dy$$

$$= \int f(y)\, g(x-y)\, dy + \int f(y)\, h(x-y)\, dy$$

$$= f*g + f*h$$

$$\Rightarrow f*(g+h) = f*g + f*h \quad \checkmark$$

4.

The Fourier transforms a signal from original domain into the frequency domain.

$$\widehat{f*g}(\xi) = \int\int f(y)\, g(x-y)\, e^{-2\pi i x\cdot\xi}\, dy\, dx$$

$$z = x-y \Rightarrow x = y+z$$

$$\widehat{f*g}(\xi) = \iint f(y)\, g(z)\, e^{-2\pi i (z+y)\cdot\xi}\, dz\, dy$$

$$e^{-2\pi i (x+y)\cdot\xi} = e^{-2\pi i y\xi} * e^{-2\pi i z\cdot\xi}$$

$$\widehat{f*g}(\xi) = \underbrace{\int f(y)\, e^{-2\pi i y\xi}\, dy}_{\hat{f}(\xi)} \underbrace{\int g(z)\, e^{-2\pi i z\cdot\xi}\, dz}_{\hat{g}(\xi)}$$

$$\Rightarrow \widehat{f*g}(\xi) = \hat{f}\,\hat{g} \quad \checkmark$$

5.

$$\langle f, h\rangle = \sum_j f_j \overline{h_j} \quad \Leftarrow \text{Adjoint}$$

$$\langle Gf, h\rangle = \sum_j \sum_i f_i\, g_{j-i}\, \overline{h_j}$$

$$\xrightarrow{m=j-i} \sum_i f_i \left(\sum_m \overline{g_m}\, \overline{h_{i+m}}\right)$$

$$\xrightarrow[\text{adjoint?}]{\text{Compare to}} \langle Gf, h\rangle = \langle f, G^* h\rangle$$

$$(G^* h)_i = \sum_m \overline{g_m}\, h_{i+m}$$

$$= \sum_j h_j \overline{g_{j-i}} = (h * \tilde{g})_i$$

$$\Rightarrow \tilde{g}_k = \overline{g_{-k}} \quad \checkmark$$